

MOHD SOHAIL KHAN

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Summary

Java Backend Engineer with **2.5 years** of production experience at **Jio Platforms** (400M+ users) — sole backend engineer on the team, owning architecture decisions end-to-end across **distributed systems**, **high-availability microservices**, and **event-driven pipelines** processing **150,000+ events/hour**. Deep hands-on expertise in **Apache Kafka**, **Spring Boot**, **PostgreSQL**, **Hazelcast**, and **Elasticsearch**. Currently exploring **AI/LLM integration** in Java backend systems — building toward production-grade intelligent services using **OpenAI** and **Anthropic APIs**.

Technical Skills

Languages: Java (8, 11, 17, 21), Python, JavaScript, SQL

Backend: Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate (ORM), RESTful APIs, OAuth 2.0, JWT, Microservices, High Availability, Fault Tolerance, Domain-Driven Design (DDD), AES/RSA Encryption

Messaging & Streaming: Apache Kafka, Spring Kafka, Event-Driven Architecture, Kafka Producer/Consumer API, Dead-Letter Queues (DLQ), Async Processing

Search & Cache: Elasticsearch, Full-Text Search, Geo-Spatial Search, Hazelcast (Distributed Cache), Redis, Query DSL

Databases: PostgreSQL, Oracle DB, Query Optimization, Indexing, Partitioning, Schema Design

Cloud & DevOps: AWS (S3, EC2), Docker, Nginx, API Gateway, Jenkins, CI/CD Pipelines, Git, Maven

AI (Exploring): OpenAI API (GPT-4o), Anthropic Claude API, Prompt Engineering, RAG concepts, Vector Search, SSE Streaming

Testing & Design: JUnit 5, Mockito, JMeter, TDD, System Design, Distributed Systems, SOLID Principles **Frontend:** React.js (basic), Next.js (basic), HTML5, CSS3 **Tools & IDE:** IntelliJ IDEA, Cursor, STS Postman, Swagger/OpenAPI, Jira, GitHub Copilot

Experience

Jio Platforms Limited

Software Development Engineer I — Location Based Services

Dec 2023 – Present

Mumbai, Maharashtra

- Owned architecture and delivery of **15+ high-availability Spring Boot microservices** (25+ production instances) as the **sole backend engineer** on the team; made all system design decisions covering service decomposition, data modeling, and infrastructure — achieving **99.9% uptime** under sustained high-concurrency load.
- Designed and scaled **Apache Kafka event-driven pipeline** (Spring Kafka) to **150,000+ messages/hour**; reduced inter-service latency by **30%** and eliminated cascading failures by decoupling producers from consumers via async processing and **DLQ-based** error recovery.
- Integrated **Elasticsearch** for full-text and geo-spatial search, replacing slow PostgreSQL LIKE scans with Query DSL on optimized index mappings — achieving **65% faster search response time** across high-cardinality location datasets.
- Drove **PostgreSQL performance** across **50M+ records** via range partitioning and composite indexing (**40% faster queries**, **28% lower CPU**); layered **Hazelcast** distributed caching to cut redundant DB round-trips by **35%** on high-frequency read paths.
- Hardened **API security layer: OAuth 2.0 + JWT** via Spring Security, **Nginx** reverse-proxy routing, and **AES/RSA payload encryption** for location data in transit — enforcing field-level security without impacting P99 latency.
- Delivered **RetailVista demographic analytics APIs** managing **22,000+ JioMart/Ajio retail assets**, powering store placement intelligence behind **10-minute delivery** operations nationwide.

Projects

Transaction Notification Service

Java, Spring Boot, Kafka/JMS, PostgreSQL, JWT/OAuth, Async

- Built event-driven financial payment processing and notification service handling **10,000+ daily payment events** using **Kafka** and **asynchronous processing**, improving system responsiveness by **25%**; implemented **JWT based authentication and role-based authorization** ensuring secure transaction workflows.
- Developed **high-performance batch export pipeline** for **1M+ financial transaction records** using pagination and streaming, supporting **audit, compliance, and reporting requirements** for large-scale payment systems.

Employee Workflow Platform

Java, Spring Boot, Kafka, PostgreSQL, React.js, Docker, Microservices

- Designed **Kafka distributed workflow engine** processing **50,000+ events/hour**, cutting delays by **40%**; decomposed monolith into **6 DDD microservices** with Docker, reducing release cycle by **30%**.
- Built dead-letter queue error handling pipeline, reducing integration failures by **35%** and improving overall **large-scale, high-performance system** reliability.

Education & Certifications

Chandigarh Group of Colleges

B.Tech, Information Technology | CGPA: 8.2/10

Jul 2019 – Jun 2023

Mohali, Punjab

IBM Java Developer Professional Certificate — Coursera

Nov 2025 – Mar 2026

Software Engineering Simulations — JPMorgan Chase, HPE, Wells Fargo (Forage)

Jan 2026